Process Evaluation

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Process Evaluation (A11523)

Short Title:	Process Evaluation		
Faculty	Science and Computing		
Department	Science		
Cluster	Mathematics and Physics		
Author	FLEONARD		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

This module will expose the student to further control chart constructs. It will also introduce the student to the principles and practice of experimental design using a suitable statistical package such as Minitab. The student will learn how to identify the appropriate design type for introductory problems, correctly conduct the experiments, input the data in to Minitab, perform the appropriate analysis and properly interpret the output.

Programmes

EVAL-0003	BSc in Good Manufacturing Practice and Technology (WD_SGMPT_D)	1 / 3 / M

Indicative Content

- Introduction to Minitab: loading datasets; obtaining summary statistics; generating charts (simple, multiple-clustered); identifying and locating outliers; data subset selection; deriving/calculating new variables
- Statistical process control (SPC): control charts for variables (mean, cusum, S, range); control charts for attributes (np, p, c, u); identifying out of control signals
- Introduction to statistical inference: two sample t-test (independent and paired). Conducting the analysis and interpreting the output - confidence intervals v p values; testing assumptions; non-parametric tests; reporting findings
- Experimental design: terminology - experimental units, factors (design/nuisance/etc.), levels, randomisation, replication, blocking; completely randomised designs; randomised block design; experimental design in Minitab; one-way ANOVA and assumptions; post-hoc analysis.
- Factorial designs: main and interaction effects - estimation and visual representation
- Advanced topics: fractional factorial design; nested factors; split plot
- Regression: regression applications in experimental design; interpreting regression analysis output from Minitab

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Construct variable and attribute control charts.
2. Compute summary statistics using Minitab.
3. Analyse comprehensively a statistical inference involving the difference in means between two groups.
4. Apply ANOVA and critically interpret the analysis.
5. Communicate the principles and the terminology of experimental design.
6. Illustrate possible application areas for some advanced concepts in experimental design, e.g. interaction effects, random factors, split plot, regression models.

Learning and Teaching Methods

- Lectures.
- Computer-based practicals.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	100%	1,2,3,4,5,6

Assessment Criteria

<40%: Unable to apply experimental design techniques to simple problems or construct control charts.

40%-49%: Able to apply experimental design techniques to some simple problems and to construct some control charts.

50%-59%: Able to apply experimental design techniques to a range of problems and to construct a comprehensive range of control charts.

60%-69%: As above and able to do some simple manipulation with advanced designs.

70%-100%: As above and able to do extensive manipulation with advanced designs.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	0	24
Independent Learning	0	111

Essential Material

- "WIT." WIT Moodle page. http://moodle.wit.ie/

Supplementary Material

- Doty, L.A. _Statistical Process Control_. New York: Industrial Press, 1996.
- Montgomery, D.C. _Design and Analysis of Experiments_. 8. USA: Wiley, 2013.
- Murdoch, J. and J.A. Barnes. _Statistical Tables_. 4th Edition. London: Macmillan, 1998.
- Reilly, J. _Using Statistics_. Ireland: Gill and McMillian, 2006.

Requested Resources

- Room Type: Computer Lab